

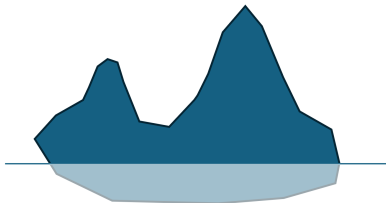
New method and MATSim contrib for the Agent-based Traffic Assignment Problem (ATAP)

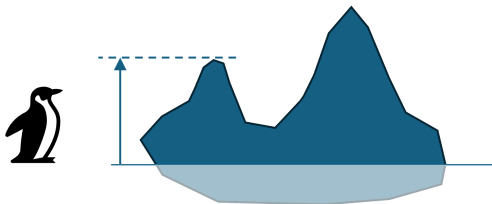
Gunnar Flötteröd
September 28, 2026

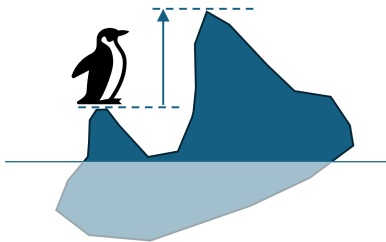


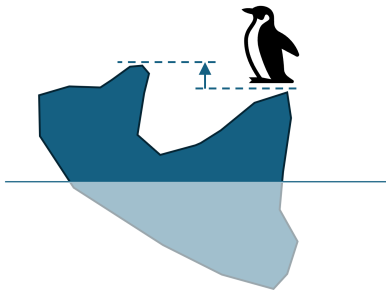
Nightmare article, started in 2018, published in 2026

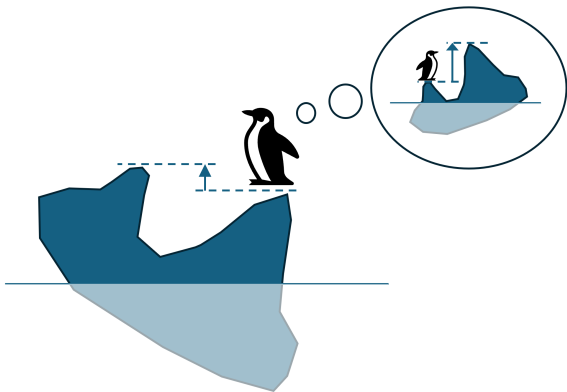


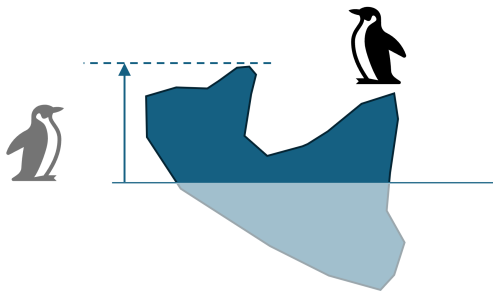


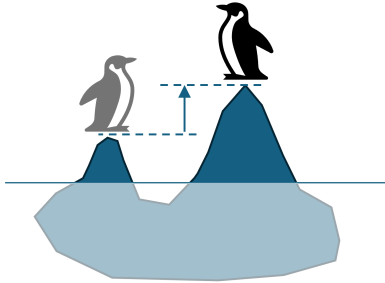


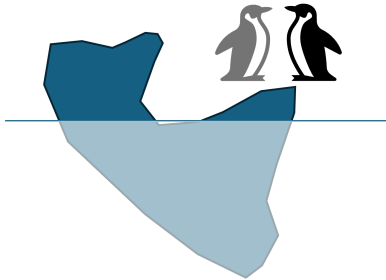












Co-evolution

- Each agent reacts individually to its current situation.
- Long-term state is result of uncoordinated agent interactions.

Assignment

- Start by articulating the desired result: Minimal agent disappointment.
- Assign agents to new positions, try to anticipate their interactions.

Assignment

Simulation process

1. Distribute all agents arbitrarily in the system.
2. Repeat many times:
 - 2.1 Let each agent decide what it wants to do.
 - 2.2 Decide which agents get to switch their behavior.

Criterion to select switching agents

- Prefer agents that currently experience large disappointments.
- Switch agents that do not affect all other agents too much.

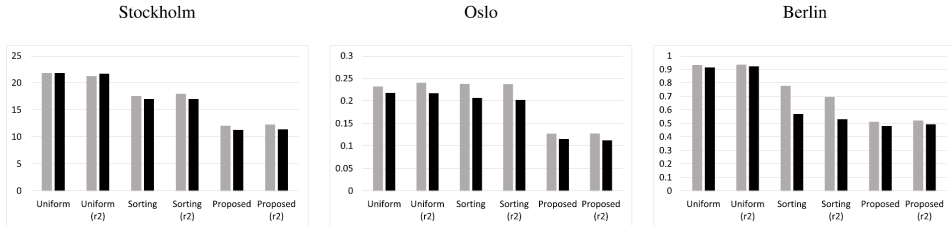
Selection of plan switchers

- Agent n expects that switching from plan x_n to y_n improves score by $\Delta S_n(x_n \rightarrow y_n)$.
- $\gamma \geq 0$ is a control parameter that expresses what we hope to achieve. $\gamma \xrightarrow{\text{many iterations}} 0$
- Switching plans from $x = (x_n)$ to y changes the travel times z by ca. $\|z(x) - z(y)\|$.
- Switch to plans $x^* = (x_n^*)$ that satisfy:

$$x^* \in \arg \max_y \frac{\sum_n \Delta S_n(x_n \rightarrow y_n) - \gamma}{\|z(x) - z(y)\| + \varepsilon}.$$

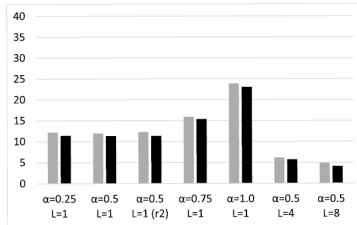
- Gets large if:
 - ▶ Numerator gets large, i.e. if the more disappointed agents get to switch;
 - ▶ denominator gets small, i.e. if switching does not affect other agents too much.
- Formulation has been derived to yield the most careful switching strategy for any γ .

Results



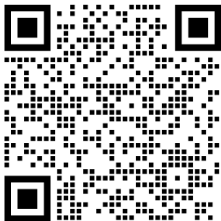
- Bars represent agent disappointment (measures deviation from equilibrium).
- “Uniform” $\approx$ “Simple recipe” from before + PSim-like replanning.

Results



- $\alpha = 0.5$ corresponds to “slow cooling”.
- $L = 1, 4, 8$ is number of iterations over which scores are averaged.

ATAP Contrib



Article

